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Original Article

cLDM-ODE: A Multimodal Generative Framework for Uncertainty-Aware Forecasting of Alzheimer's Disease Progression

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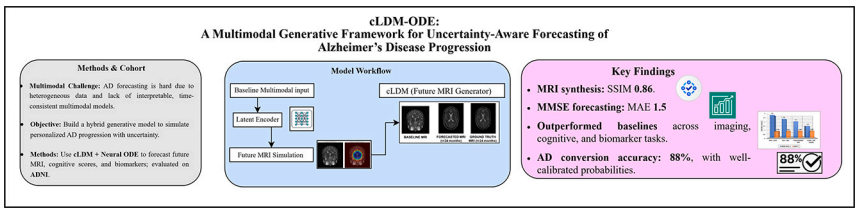
HIGHLIGHTS

- A generative model forecasts MRI, cognition, and biomarkers over time.
- Integrates diffusion, ODEs, and classification in a unified pipeline.
- Uses multimodal inputs: MRI, MMSE, ADAS-13, Aβ, and tau biomarkers.
- Achieves state-of-the-art accuracy in AD conversion and score prediction.
- Supports uncertainty-aware, personalized disease trajectory simulation.

GRAPHICAL ABSTRACT



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ABSTRACT

Context: The imminent performance of multimodal and heterogeneous modalities to forecast the advance of Alzheimer's disease (AD) precisely is one of the key problems. Current models are usually not interpretable, time-consistent, and multimodal, making them less useful in clinical forecasting.
Objective: The objective of the study is to develop a hybrid generative approach to simulate the individualized AD progression process, which can generate future anatomical and clinical states, model latent over-time dynamics, and measure the uncertainty.
Methods: The proposed study suggests using a multimodal paradigm that enables a combination of Conditional Latent Diffusion Models (cLDM) and Neural Ordinary Differential Equations (ODEs). The model permits the generation of plausible future MRI, cognitive scoring, and biomarker trajectories for a patient at baseline. The ADNI dataset was evaluated with structural similarity (SSIM), clinical prediction error, and classification accuracy.
Key Findings: The model provided an SSIM equal to 0.86 on synthesizing future MRI, and the MAE of MMSE prediction was equal to 1.5. It exceeded baselines in all the imaging, cognitive, and biomarker settings. The conversion of AD resulted in an accuracy of the classification of 88% with stable multimodal generalization at calibrated output of probability.
Conclusion: The proposed model offers a feasible and explainable approach to the forecast of an AD trajectory, allowing realistic simulations of a digital twin and projecting its progress within a multi-year perspective. It also supports early detection, custom intervention, and uncertainty-conscious clinical decision-making.
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1. Introduction and background

AD is a long-term, ongoing neurodegenerative disease that mostly affects older people. The condition features the buildup of amyloid-beta (Aβ) plaques, tau protein tangles, and the death of many neurons, all of which cause permanent brain shrinkage [1]. Clinically, AD is recognized by a slow decline in thinking, mem-

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Table 1
Comparative analysis across AD progression stages

Feature category	CN	MCI	Alzheimer's dementia
Clinical symptom	Normal cognition	Episodic memory loss	Multi-domain decline
Functional status	Independent	Mostly independent	Dependent on assistance
MMSE score range	28-30 [3]	24-27 [4]	≤23 [4]
Hippocampal volume	Normal	∼10- 15% reduction [5]	>20% reduction [5]
CSF $A\beta$ levels	Normal	Decreasing trend [4]	Significantly reduced [5]
Tau protein levels	Normal	Elevated [4]	High [5]
Conversion risk	Low	∼10- 15% year [4]	N/A
Modeling challenge	Baseline	High inter-patient variability	Late-stage degeneration

ory, language, and decision-making, which often begins mildly and worsens over time [2]. Mild cognitive impairment (MCI) acts as a crucial stage, with patients having noticeable cognitive problems but still being able to function on their own. Because not every MCI patient develops Alzheimer's dementia, it makes it difficult for doctors to diagnose and predict the future [3]. What makes things more difficult is that disease onset and how quickly it progresses vary a lot among people, due to genetic, demographic, and biomarker differences [4]. For example, even though hippocampal atrophy is a well-known sign of early AD, it progresses at different rates in different people, and cognitive problems do not always occur at the same time as these brain changes [5]. As of 2023, more than 55 million people worldwide have AD, and by 2050, this could rise to 139 million, putting great pressure on public healthcare systems [6]. These numbers clarify that we need better predictive tools to track how neurodegeneration will progress over time and space. In particular, the model should predict when someone with MCI is expected to develop Alzheimer's dementia [7]. It should also be able to predict future changes in brain structure, such as bigger ventricles or a smaller hippocampus, which signal neurodegeneration. As well as predicting brain changes, the model should also show how cognitive scores, like those from the mini-mental state examination (MMSE) or AD's assessment scale (ADAS-13), might change over time. These predictions must be made using a probabilistic system that includes uncertainty, so clinicians can evaluate risk in several potential disease paths [8]. Traditional diagnostic pipelines use cross-sectional evaluations or primitive longitudinal approaches such as linear regression or survival analysis that do not reflect the nonlinear, dynamic, and high-dimensional nature of neurodegeneration [7]. Therefore, there is a rising agreement that temporal models based on deep learning, especially generative architectures, have the potential to transform AD prediction from reactive evaluation to proactive prediction. Table 1 compares the clinical, anatomical, and biomarker features at the three main stages of AD: cognitively normal (CN), mild cognitive impairment, and Alzheimer's dementia. This table points out that disease progression is complex and nonlinear, as MMSE scores, hippocampal volume, and CSF biomarker levels change at different rates and times. Although early AD can be detected by shrinking hippocampal volume and amyloid-beta buildup [5], the actual appearance of cognitive problems is often delayed, which makes forecasting the disease difficult [4]. In terms of computational modeling, the MCI stage is hardest to tackle because patients are very different and the chance of conversion is uncertain, ranging from 10% to 15% yearly based on biomarkers and imaging [4]. This makes it even more important to use the proposed generative forecasting as it is a probabilistic, multimodal, and trajectory-aware model approach.

1.1. Motivation

Despite rapid advancements in medical imaging and machine learning (ML), most computer models that predict AD still focus on making one-time diagnoses. These models are usually trained to classify brain scans from one time and figure out if the person is CN, has MCI, or has Alzheimer's dementia based on structural MRI or PET images. DL frameworks like Convolutional Neural Networks (CNN) have shown their capability in accurately predicting AD from brain scans [9], [10]. For instance, studies by Liu et al. [9] and Basaia et al. [11] have found that using 3D CNNs trained on baseline MRI scans can distinguish patients with normal cognition from those with AD with an accuracy of over 90%. However, these methods have a significant limitation in handling a specific type of data. They analyse each piece of information as a single, one-time snapshot, without considering how the disease might change over time [12]. In contrast, Alzheimer's is a disease that gets worse over time, and it can take years or even decades for symptoms to show up and get worse in different people. Thus, there is a need for a novel approach that focuses on forecasting rather than just diagnosing AD, and the model should be able to predict how the disease might progress in each individual and generate personalized information. The proposed study suggests that AD should be seen not as a fixed label, but as a disease that changes and grows in each person. Such a perspective employs the researchers as well as clinical doctors not only to analyse how the brain might change over time but also to predict how fast a patient's thinking might slow down and when they might be at higher risk for problems. This changes the perspective of AI in AD and opens up the idea of using patient data to create virtual models of the disease, which can help doctors analyze how the disease might progress in different situations.

Table 2 presents a comparative analysis between conventional static classifiers and the proposed temporal forecasting framework based on conditional latent diffusion and latent dynamics modeling. This contrast illustrates the conceptual and practical evolution of computational AD modeling from diagnosis-based systems to forecasting-oriented architectures.

1.2. Limitations of current longitudinal models

While longitudinal models represent a step forward from static classifiers in AD research, many are limited in both predictive capacity and clinical relevance [13]. While time-aware models like RNNs and LSTMs have been used to forecast scalar outcomes like MMSE scores over time, these approaches cannot capture the high-dimensional anatomical changes in the brain, such as structural atrophy or ventricle enlargement. In addition, most existing models work in the score space only and provide no anatomical visualization to aid interpretability [14]. Generative methods like GANs and VAEs have been proposed for synthesizing neuroimages, but they typically do not support multi-step temporal forecasting or multimodal inputs or uncertainty estimation, all of which are necessary for clinical decision making. A persistent problem is the cognitive–anatomical prediction disconnection, with most frameworks describing either cognitive decline or brain degeneration, but not both in a single framework. Moreover, probabilistic modeling is often missing, which makes it difficult to do confidence calibration in real-world applications [15]. Table 3 provides a com-

Table 2
Comparison Between Static and Temporal Forecasting Models for AD

Feature /Capability	Static Classifier	Temporal Forecasting Model
Input Type	Single-time-point MRI	Longitudinal baseline + optional multi-modal inputs
Output	Diagnosis label	Forecasted future MRI, MMSE, and conversion time.
Time-Awareness	× Absent	✓ Explicitly modeled
Trajectory Simulation	× Not possible	✓ Can simulate future states
Conversion Time Estimation	× No	✓ Yes
Multimodal Integration	Partial or absent	✓ Biomarker+ MMSE + imaging
Uncertainty Quantification	× Typically, absent	✓ Included
Clinical Trial Utility	Limited (diagnosis only)	High (timeline forecasting, digital twin modeling)

Table 3
Comparison of Existing Longitudinal Models in AD Forecasting

Study	Model Type	Modality	Temporal Modeling	Forecasting	Output Type	Uncertainty	Computational Complexity	Deployment Difficulty
[9]	Multi-model CNN	Structural MRI	×	×	Class + Segmentation	×	Low	Easy
[12]	Latent Feature-Enhanced GAN (LFGAN4Tau)	Tau PET	×	✓	Image (tau deposition map) + Risk Score	×	Moderate	Moderate
[16]	Self-Supervised Deep Latent Model with Clustering	Structural MRI	✓	✓	Latent progression patterns + Subtypes	×	Moderate	Moderate
[17]	Neural Processes + Transformer Encoder	Longitudinal clinical data	✓	×	Diagnosis class + Disease Progression	✓	Moderate	Moderate
[18]	Feature Tagger + TimesBlock + Cross-modal Attention	EEG + MRI + Clinical	✓	×	Diagnostic class	×	Moderate to High	Moderate
[19]	Generative Feature Extractor + Mamba Block	MRI + PET + Assessment Scales	✓	✓	MCI-to-AD Progression + Risk Prediction	×	Moderate to High	Moderate
[20]	Hybrid Spiking Neural Network (SNN)	3D MRI	×	×	Diagnosis Class	✓	Low	Easy
Proposed Study	**cLDM + ODE + Classifier**	**MRI + Biomarkers + MMSE**	✓	✓	**Image + Score + Timeline**	✓	**High**	**Moderate to Difficult**

parative overview of key recent studies in longitudinal AD forecasting to highlight these gaps.

As shown in Table 3, a majority of the longitudinal models of AD that have been developed have narrowed down to neuroimaging and cognitive data. Nevertheless, the temporal modeling of diffusion is becoming more widespread in healthcare fields. As Zhong et al. suggested, MedDiffusion aims to augment the prediction of health risks based on multimodal EHR data through diffusion processes [21]. Naseer et al. proposed ScoEHR, a system that uses continuous-time diffusion to develop realistic synthetic EHR time series [22]. Outside of the healthcare sector, Wen et al. created DiffSTG, which is a spatio-temporal diffusion framework used to forecast uncertainty in dynamical networks [23]. These studies illustrate that the application of diffusion models integrated with temporal dynamics is becoming a common practice and not confined to the field of neuroimaging, which gives additional context to the proposed cLDM-ODE framework within a broader trend of temporal generative modeling in the clinic and real-world applications.

1.3. The need for generative forecasting

Although temporal modeling has progressed, most existing frameworks are still only capable of classification or scalar score prediction and provide little insight into the anatomical progression or uncertainty of AD [24]. To overcome this, the current work has proposed a hybrid generative forecasting model that learns the distribution of neurodegenerative patterns from longitudinal brain imaging and clinical data. In contrast to deterministic models, the proposed approach produces plausible future MRIs and cognitive scores given baseline states, modeling structural and functional evolution over time [25]. The model can simulate multiple potential disease trajectories by modeling latent disease dynamics as components of a conditional latent diffusion framework, using neural ODEs or GRUs [26]. The probabilistic forecasting approach

presented here enables the creation of a "digital twin" – a virtual replica of the patient that evolves in silico and allows for personalized prediction, risk stratification, and scenario planning for clinicians and trial designers. As such, generative forecasting provides powerful tools for closing the gap between static diagnosis and dynamic, interpretable disease modeling.

1.4. Contribution

- In this proposed study, a new deep generative architecture is introduced that combines a cLDM with Latent Dynamics Modeling and a Classifier Head to jointly simulate and predict neurodegeneration and clinical conversion timelines.
- The proposed work builds a model that integrates heterogeneous input data (structural MRI, cognitive scores, and CSF biomarkers (Aβ, tau)) to capture the multifaceted progression of AD and enhance prediction robustness.
- The framework allows for the creation of in silico simulations of disease evolution for individualized, interpretable, and temporally resolved forecasts of brain atrophy and cognitive decline.
- The model incorporates probabilistic generative modeling through the diffusion process, enabling uncertainty quantification and multi-scenario planning, thus making the model more applicable to real-world clinical decision making.
- The proposed system is rigorously evaluated on the ADNI dataset using subject-wise longitudinal splits. The performance is assessed via SSIM metrics, MMSE forecasting error, conversion classification accuracy, and calibration reliability for a range of forecasting tasks.

In contrast to existing literature, the novelty of the proposed model is in the fact that it combines conditional latent diffusion with continuous-time neural ODEs, and as a result, this offers an end-to-end and uncertainty-aware solution to a longitudinal multimodal forecasting task. Unlike previous research that is either concerned with predicting the scalar scores or generating the MRI, the proposed is the first to jointly predict the future of the MRI scans, cognitive outcomes, and biomarker profiles within a clinically-deployable digital twin framework that can be interpretable. This makes the proposed architecture a holistic improvement of the baseline generative/temporal architecture.

2. Problem formulation and theoretical foundation

AD progression is a complex, nonlinear, and multimodal process that includes anatomical degeneration, biochemical disruption, and cognitive decline over time. However, traditional machine learning models find it difficult to capture the complexity of this problem, especially when asked to produce individualized forecasts that are both clinically realistic and uncertainty-aware [27]. To tackle this challenge, we propose a framework that leverages principles from probabilistic generative modeling, latent variable theory, and neural differential equations, which serves as the basis for a hybrid deep learning model for in silico simulation of disease evolution.

2.1. Theoretical foundation

The core of the proposed work is generative modeling under uncertainty, where the model does not predict a single future outcome, but instead learns a distribution over possible future trajectories given the current patient state. In generative modeling, specifically in latent variable methods like Variational Autoencoders (VAEs) and Diffusion Probabilistic Models, we want to learn the data distribution $p(X)$ by introducing a lower-dimensional latent space Z defined by equation (1).

$$p(X) = \int P(X \mid Z)\, p(Z)\, dZ \tag{1}$$

In the conditional setting, the modeling was done for temporal prediction, as shown in equation (2).

$$p(X_{t+T} \mid Z, X_t, T)\, p(Z \mid X_t, T)\, dZ \tag{2}$$

However, the proposed study uses diffusion-based learning to learn these distributions flexibly and accurately, unlike classical latent variable models, which assume simple priors and tractable posteriors [28]. Furthermore, we treat disease progression as a continuous-time dynamical system in latent space, which is mathematically represented as shown in equation (3).

$$\frac{dz(t)}{dt} = f_{\emptyset}(z(t), t) \tag{3}$$

This modeling perspective is consistent with biological intuition that disease progression is continuous, smooth, and individualized rather than stepwise or discrete.

2.2. Problem definition

Patient's state at time t is a tuple composed of multimodal observations as shown in equation (4):

$$X_t = \{I_t, C_t, B_t\} \tag{4}$$

Where I_t, C_t and B_t represents the structural brain image, cognitive score vector, and biomarker vector, respectively. These are the structural, functional, and biochemical states of the brain, and their primary task is to model and generate future states represented as $\hat{X}_{t+T}$ where $T \in \mathbb{R}^+$ represents the time offset into the future. Given a current patient state and a forecast horizon, we define a forecasting function G_θ parameterized by θ that maps to a distribution over future outcomes as shown in equation (5).

$$\hat{X}_{t+T} = G_\theta(X_t, T) \tag{5}$$

The decomposition into individual targets of the above predictive mapping has been shown in equation (6).

$$\hat{X}_{t+T} = \{\hat{I}_{t+T}, \hat{C}_{t+T}, \hat{B}_{t+T}, \hat{y}_{t+T}\} \tag{6}$$

Where $\hat{I}_{t+T}$, $\hat{C}_{t+T}$, $\hat{B}_{t+T}$ and $\hat{y}_{t+T}$ denotes the generated future MRI capturing anatomical change, predicted cognitive scores, predicted biomarker levels, and estimated probability of AD conversion, respectively. Equations (4) to (6) show a typical case of conditional temporal generative modeling in multimodal medical time series. In the current case, we do not try to directly predict a specific outcome in the future. We focus on learning how to move from the current state to a range of possible future states, which allows us to forecast disease progression in both certain and uncertain ways.

2.3. Conditional latent diffusion modeling

In the proposed work, cLDM is used to simulate and forecast AD progression over time, producing plausible future states of patient-specific neuroimaging and clinical profiles. Score-based generative models such as diffusion models have recently demonstrated strong promise in generating high-fidelity, diverse, and uncertainty-aware outputs across medical domains [29]. Diffusion models learn a data distribution p(X) by training a model to reverse a forward noising process that gradually perturbs a data sample into Gaussian noise. The purpose of the study is to reduce the complexity of training and sampling by learning a diffusion

process in a simpler latent space for 3D MRIs. Let Z_0 be the latent representation of the future patient state $\hat{X}_{t+T}$ generated using an encoder. The forward process corrupts Z_0 via a sequence of noisy latent z_t, which uses a predefined variance schedule $\alpha_t \in (0, 1)$, as shown in equation (7).

$$q(z_t \mid z_0) = N(z_t; \sqrt{\alpha_t} z_0, (1 - \alpha_t) I) \tag{7}$$

The process shown in equation (7) incrementally adds Gaussian noise to z_0, where $t \in \{1, 2, \ldots\ldots, T\}$ indicates the diffusion time steps. Parameterized by a neural network p_θ, the reverse process tries to denoise z_t given the current noisy state and an external context $c = \{X_t, T\}$, where X_t and T denotes the current patient's multimodal state and forecast horizon.

2.4. Latent dynamics as continuous-time process

To simulate the temporal progression of AD effectively, the proposed study embeds patient-specific dynamics in a continuous-time latent space such that the model evolves the encoded state of a patient smoothly over arbitrary time horizons [30]. This is important for personalized forecasting, especially in cases where patient follow-up intervals differ or data is missing at intermediate stages. Rather than relying only on discrete step transitions, we adopt a continuous dynamical system governed by Neural ODEs to model disease evolution. With this latent embedding in hand, we define its temporal evolution via a differential equation parameterized by a neural function $f_\emptyset$ as shown in equation (8).

$$\frac{dz(s)}{ds} = f_\emptyset(z(s), s) \tag{8}$$

Where $s \in [t, t + T]$ denotes continuous time, and $f_\emptyset$ controls the variations of the latent state based on its current value and time. As shown in equation (9), the future latent state at time $t + T$, which is the evolved patient condition, is computed by integrating over the dynamic process.

$$z_{t+T} = z_t + \int_t^{t+T} f_\emptyset(z(s), s) ds \tag{9}$$

This continuous time formulation enables the current study to generate future latent states at arbitrary horizons $T \in R^+$ even if the dataset does not contain fixed interval observations. This flexibility is especially useful in AD forecasting, where progression is slow, non-linear, and patient-specific [31].

2.5. Multi-objective optimization

The proposed work ensures that multiple important outcomes in AD progression, such as changes in the brain, cognitive loss, shifts in biomarkers, and prediction of conversion, are learned together by using a multi-objective optimization approach [32]. This approach helps the model to improve each part of the forecasting task by working on several learning goals at once. Every component adds its loss function, which represents a different way of predicting, and all of them are combined and weighted to create the global training goal [33]. The proposed study constructs the total loss function L_{total} as a weighted sum of 4 primary components: diffusion loss from the generative imaging model, latent dynamics consistency loss from the neural ODE-based temporal modeling, regression loss for cognitive and biomarker forecasting, and classification loss for predicting conversion from MCI to AD. This can be expressed as demonstrated in equation (10).

$$L_{total} = \lambda_1 L_{diff} + \lambda_2 L_{latent} + \lambda_3 L_{reg} + \lambda_4 L_{cls} \tag{10}$$

Where L_{diff}, L_{latent}, L_{reg} and L_{cls} represents the diffusion loss, latent dynamic loss, regression loss, and classification loss. And the balancing weights λ_1, λ_2, λ_3 and $\lambda_4 \in \mathbb{R}^+$ ensures the trade-off between the competing objectives. The choice of these hyperparameters is based on validation performance and clinical importance, like giving higher weight to diffusion loss when anatomical realism is valued more than classification loss when conversion prediction is the main goal. To prevent the model from overfitting to a single task, this composite optimization strategy optimizes the model to learn a shared latent representation that can simultaneously support imaging synthesis, clinical forecasting, and diagnostic risk estimation [34]. In real-world Alzheimer's applications, such unified learning is crucial to reconcile multimodal evidence to inform patient-specific decisions. This improves model interpretability and robustness and provides a basis for uncertainty-aware, digital twin-based planning in AD clinical pathways.

3. Methodology

The methodology is based on combining neuroimaging, cognitive scores, and biochemical biomarkers to project future conditions of patients. The framework depends on three main modules: a CLDM for predicting anatomy, a latent dynamics module for time-based modeling, and a classification head for predicting AD conversion. All the elements are part of a multi-objective optimization that allows the system to forecast neurodegeneration, cognitive decline, and the transition to a diagnosis at the same time.

3.1. Overview of the pipeline

A unified methodology is used, which involves using both generative and temporal learning modules to model the progression of AD. It uses baseline MRI, cognitive scores, and CSF biomarkers to create a shared latent space with the help of a variational encoder. After that, the latent representation goes through two parallel paths: a model that creates detailed future brain MRIs and a module that models the ongoing disease progression in the patient's latent space. By interpreting the hidden state, the model estimates the chances of cognitive decline, biomarker changes, and a person shifting from MCI to AD. All these outputs are monitored using specific loss functions in a multi-objective framework to ensure that learning about anatomy, prediction, and diagnosis occurs together. Because of the system, digital twin simulations can be run for individual patients and show how different diseases develop over any chosen period. Fig. 1 shows the overview of the complete methodology pipeline. The modules are also connected via a shared latent representation. Additionally, the conditional latent diffusion model is used to encode the multimodal inputs as a denoised latent state, which in turn evolves using a latent ODE solver. These trajectories are decoded at a later stage to produce MRI data and fed to task-specific heads to forecast cognitive outcomes, biomarkers, and conversion events. This interconnection ensures the unified latent space modeling of temporal dynamics, anatomical structures, and clinical signals.

Additionally, the functional components of the proposed forecasting framework are outlined in Table 4, which details how the framework processes multimodal inputs, models latent dynamics, and generates clinically meaningful outputs. This demonstrates how each module transforms input representations and generates future neuroimages, cognitive scores, and disease progression risk to form a unified and interpretable prediction pipeline for AD.

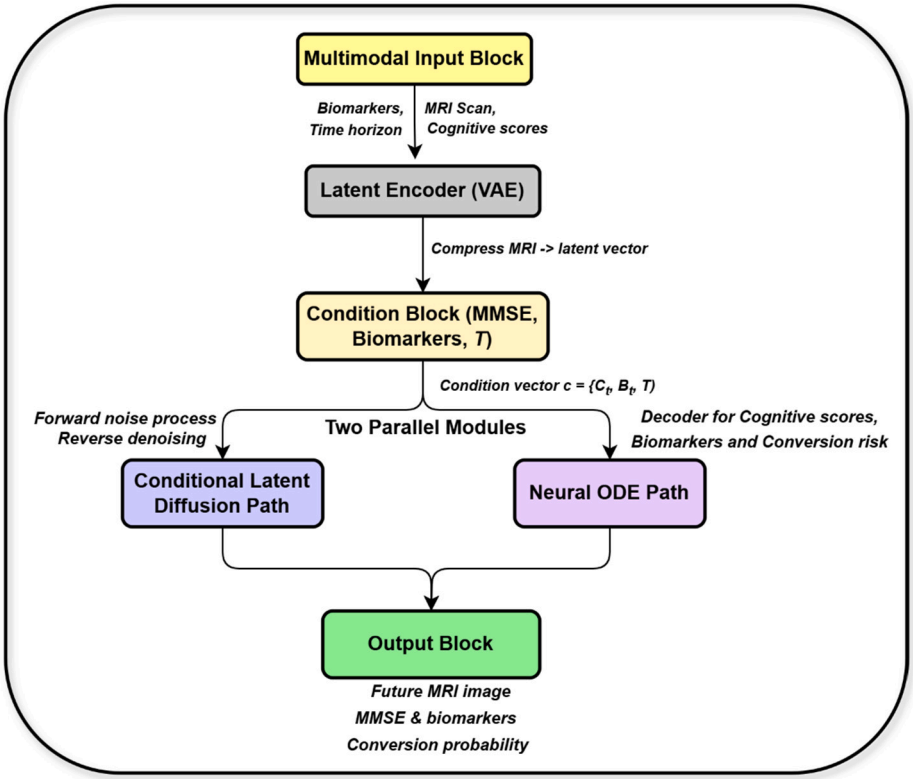

Fig. 1. Methodology overview.

Table 4
Functional Role and Output of Methodology Components

Component	Represents	Transformation Function	Output
Multimodal Input Block	ADNI dataset: MRI, cognitive scores, biomarkers, and forecast horizon	Input mapped to a unified context $X_t = \{I_t,\ C_t,\ B_t\}$	Multimodal input vector X_t and condition $c = \{C_t, B_t, T\}$
Latent Encoder (VAE)	Encodes high-dimensional MRI into a compressed representation	$z_0 \sim N(\mu_{\emptyset}(I_t),\ \sigma^2_{\emptyset}(I_t))$	z_0 denoting anatomical features
Conditioning Block	Encodes clinical context; defines conditioning information for dynamics modules	No transformation	Condition vector employed by both the diffusion and dynamics modules
Latent Diffusion Module	Generates a realistic future MRI by modeling probabilistic anatomical evolution	• Forward process (noise injection): $q(z_t \mid z_0) = N\left(z_t; \sqrt{\alpha_t}.z_0, (1-\alpha_t).I\right)$ • Reverse process (denoising): $p_\theta(z_0 \mid z_t, c) = N(z_0; \mu_\theta(z_t, c), \sum_\theta(z_t, c))$	Generate future latent representation
Neural ODE	Smooth disease progression in the latent space over time	$z_{t+T} = z_0 + \int_t^{t+T} f_{\emptyset}(z(s),\ s)ds$	Evolved latent state
Clinical decoder	Mapping of future state to clinical variables	Linear projection: $\hat{C}_{t+T} = W_{C^{z_{t+T}}} + b_C$, *etc.*	Forecasted scores and biomarkers
Conversion Classifier	Predicts binary conversion risk	$\hat{y}_{t+T} = \sigma(W_{y^{z_{t+T}}} + b_y)$	Estimated probability of conversion
Multi-Objective Losses	Simultaneous training goals	$L_{total} = \sum \lambda_i L_i$ where $i \in \{diff,\ latent,\ reg,\ cls\}$	Backpropagated loss for module optimization

3.2. Latent encoding of structural input

Structural MRIs are high-dimensional, which makes efficient modeling difficult, particularly in a forecasting framework where future neuroimages must be synthesized from baseline scans. To overcome this, the proposed study introduces a latent encoding mechanism that compresses 3D MRI volumes into a compact latent representation that is information-rich. This step is critical to reduce the computational load and to allow downstream temporal modeling via generative diffusion and latent ODEs. For a given structural MRI input at baseline I_t, the study uses a convolutional encoder E_ϕ parameterized by weights ϕ to map the input volume into a latent space as shown in equation (11).

$$z_0 = E_{\emptyset}(I_t) \tag{11}$$

Here, $z_0 \in \mathbb{R}^d$ denotes the latent vector representation of the anatomical structure at time t, where $d \ll dim\ (I_t)$. This compression retains salient spatial features such as hippocampal volume, ventricular expansion, and cortical atrophy patterns and discards redundant spatial detail.

The proposed study structures the encoder as part of a VAE setup, where the encoding is not deterministic but rather samples from a learned posterior as shown in equation (12), to ensure the latent representation captures robust, disentangled features.

$$z_0 \sim N(\mu_{\emptyset}(I_t), \sigma^2_{\emptyset}(I_t)) \tag{12}$$

Table 5
Functional Summary of Latent MRI Encoding Module

Aspect	Description
Input	3D structural MRI scan I_t
Encoder Architecture	3D CNN-based variational encoder $E_\emptyset$
Output Latent Representation	$z_0 \in \mathbb{R}^d$, sampled from $N(\mu,\ \sigma^2)$
Loss Function	Reconstruction loss + KL divergence

The probabilistic formulation encourages generalization and makes the latent space continuous and smooth, which is useful for downstream diffusion modeling. In training, the KL divergence term between the posterior and prior $N(0,I)$ is minimized along with the reconstruction loss to encourage well-behaved latent representations. The latent representation z_0 is passed through 2 major pipelines, namely, the latent diffusion module that generates ,$\hat{z}_{t+T}$ and the neural ODE block that evolves z_0 forward in time to model disease progression continuously. The current work also includes D_ψ representing a lightweight decoder shared with the diffusion model to verify the encoding fidelity, which reconstructs the brain volume from the latent space as shown in equation (13).

$$\hat{I}_t = D_\psi(z_0) \tag{13}$$

Additionally, equation (14) shows the encoder-decoder pipeline, which is jointly trained with reconstruction and KL losses.

$$\mathcal{L}_{VAE} = E_{q_\phi(z_0|I_t)}\left[\left\|I_t - \hat{I}_t\right\|^2\right] + \beta.KL(q_\phi\,(z_0 \mid I_t) \,||\, N(0,I)) \tag{14}$$

This objective guarantees that the structural information necessary for forecasting is correctly stored in the latent embedding. Table 5 shows the summary of the structural MRI encoding method that has been used in the developed framework. It details how brain scans are reduced to low-dimensional vectors, which are more useful and simpler for later forecasting. Furthermore, it supports both anatomical reconstruction and temporal evolution modeling.

A 3D convolutional encoder transforms the high-dimensional structural MRI input into a compact latent representation, which allows for efficient downstream modeling of disease progression. The encoder shown in Fig. 2 comprises stacked Conv3D layers with global average pooling and fully connected layers to produce the mean and variance of the latent distribution. This latent vector z_0 encodes essential anatomical features and reduces spatial complexity.

3.3. Conditional latent diffusion for future MRI generation

The proposed work uses a cLDM to accurately model the brain changes seen in AD, allowing for forecasting MRI with high detail and based on the patient's current state and the forecast time frame. While pixel-space methods for 3D neuroimages are expensive to compute, the proposed model works with latent space, giving good anatomical detail and making learning efficient. The complete process is initialized by encoding the baseline structural MRI I_t into a latent vector $z_0 \in \mathbb{R}^d$ employing the encoder network. This latent variable acts as the first clean description of brain anatomy. We train the model by gradually increasing the noise in the latent representation over T steps, which is known as a forward diffusion process. This is shown mathematically in equation (15).

$$q\,(z_t \mid z_0) = N(z_t; \sqrt{\overline{\alpha}_t}.z_0, (1-\overline{\alpha}_t)\,.I) \tag{15}$$

Where $\overline{\alpha}_t = \Pi_{s=1}^{t}\alpha_s$ represents the cumulative noise scaling at timestep t, and I denote the identity covariance, modeling isotropic Gaussian noise. The suggested study trains a conditional denoising network that estimates the mean of the posterior distribution given a noisy latent input and a conditioning vector to be able to reverse this process and reconstruct a clean latent vector from its noisy version. The conditioning vector $c = \{X_t,\ T\}$ is the full multimodal patient state at baseline, including MRI, cognitive scores, biomarkers, and the desired forecast horizon T. The reverse process parameterized by θ is defined as shown in equation (16).

$$p_\theta\,(z_0 \mid z_t, c) = N(z_0; \mu_\theta\,(z_t, c)\,, \sum_\theta (z_t, c)) \tag{16}$$

Where μ_θ and $\sum_\theta$ represents the output of a neural network predicting the denoised latent vector's mean and covariance, respectively. This formulation admits sampling of a plausible latent representation $\hat{z}_0$ for future anatomical states, which accounts for uncertainty in disease progression. Finally, the denoised latent sample $\hat{z}_0$ is fed through a decoder network $D_\emptyset$ to synthesize the predicted MRI volume at time $t+T$ as shown in equation (17).

$$\hat{I}_{t+T} = D_\emptyset(\hat{z}_0) \tag{17}$$

It allows the generation of future structural images grounded in the observed baseline and biologically plausible progression patterns. Table 6 presents the architecture of the conditional latent diffusion process and explains each module in terms of its functional role, input, output, and mathematical operation to capture and communicate the functioning of the conditional latent diffusion process.

Additional information, including noise schedule and sampling protocol like utilization of a linear β-schedule on training and DDIM accelerated inference during 50 and 100 steps, has been added to Table 6 to enhance similar clarity and reproducibility.

The three particular modalities used in the proposed framework are heterogeneous, including structural MRI, cognitive scores (e.g., MMSE, ADAS), and CSF biomarkers (e.g., Aβ, tau), and a late fusion strategy is employed after modality-specific encoding. Individual encoder streams are used to process the different modalities so that domain-specific structure can be preserved. The spatial features are extracted by a convolutional encoder. The numerical temporal dependencies between the cognitive scores are preserved with a 2-layer MLP, whereas the CSF markers are transformed linearly, and then batch normalization is performed. The representations of the individual modalities are concatenated and fed through a common projection head, which will project them to the latent space and use them to model time through Neural ODE. This combination of modality-specific learning and joint representation building is balanced in this strategy. For dealing with missing modality, the system uses zero masking and modality dropout during training, summarized in Table 7.

The implementation involves zero-masking and modality dropout during training; the proposed study utilized these strategies to address the missing modalities in ADNI. Namely, the missing and stochastic dropout was imposed on MRI channels with a learned missing embedding to promote robustness, and cognitive and biomarker modalities were zero-masked. In comparison to explicit imputation strategies, the design can be trained on predictive distributions with incomplete observations, which also enhances generalization. In addition to this, the neural ODE module naturally addressed the problem of irregular follow-up intervals and modeled the progression in continuous time, without relying on uniform sampling. Significantly, the framework does not require the presence of all modalities. By means of zero-masking and modality dropout, the model is trained to predict even at baseline with one or more modalities missing. This allows the proposed framework to be applied in real-world settings where missing-modality scenarios will be available.

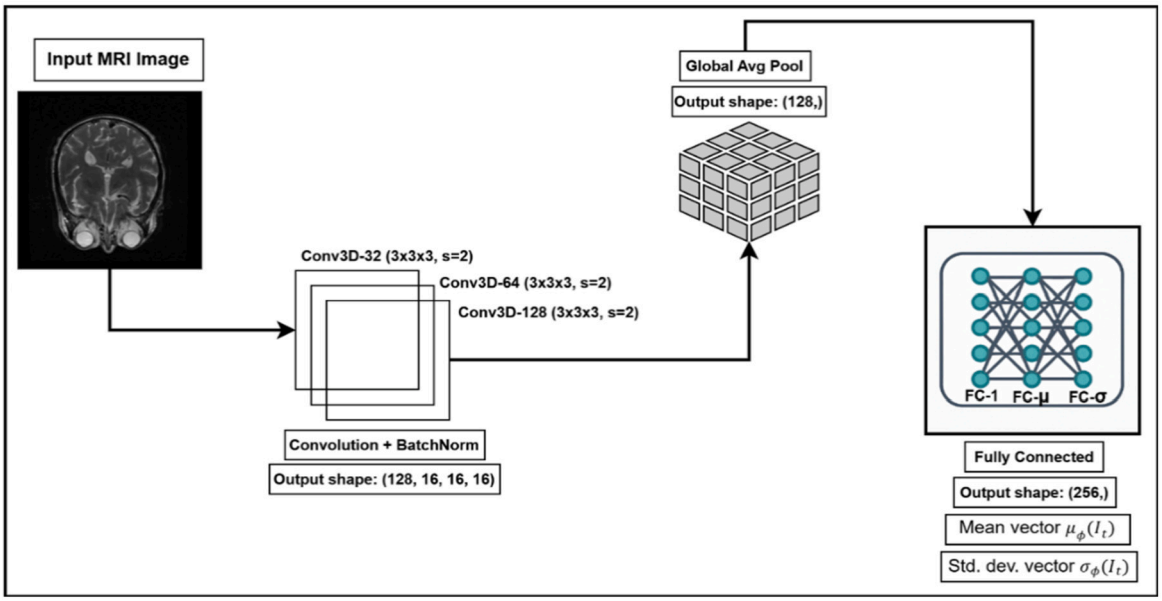

Fig. 2. Latent MRI Encoder Architecture.

Table 6
Functional Specification of Conditional Latent Diffusion Module

Component	Description	Input	Output	Operation / Mapping Function
Latent Input Sampling	Encodes baseline MRI into a compact latent representation	I_t	$z_0 \in \mathbb{R}^d$	$z_0 \sim N(\mu_\emptyset,\ \sigma_\emptyset^2)$
Forward Diffusion Process	Adds Gaussian noise to latent over T steps; Linear β-schedule from 1e-4 to 0.02	$z_0,\ t$	$z_t \in \mathbb{R}^d$	$q(z_t \mid z_0) = N(.),\ T = 1000$
Denoising UNet	Learning based on reversing the noise given the patient's state and time	$z_t,\ c = \{X_t,\ T\}$	$\epsilon_\theta \epsilon \mathbb{R}^d$	$\in_\theta (z_t, c)$
Reverse Sampling	Samples clean latent using DDIM; Fast deterministic sampling over 50-100 steps	$z_t,\ c$	$\hat{z}_0 \in \mathbb{R}^d$	$p_\theta (z_0 \mid z_t,\ c) \sim N(\mu,\ \sum)$
Latent-to-MRI Decoding	Decodes denoised latent and generates future MRI.	$\hat{z}_0$	$\hat{I}_{t+T} \in \mathbb{R}^{H \times W \times D}$	$D_\emptyset(\hat{z}_0)$

Table 7
Multimodal Fusion Configuration

Modality	Encoder	Output Dim	Missing Modality Strategy
MRI	3D CNN	128-d	Zero-masked + learned missing embedding
Cognitive Scores	2-layer MLP	32-d	Random dropout during training
CSF Biomarkers	Linear + BatchNorm	32-d	Learned noise-injected missing token
Final Fusion	Concatenation + Projection FC	192-d → latent	Shared latent projection fed to Neural ODE

3.4. Latent trajectory modeling via neural ODEs

To forecast the future trajectory of AD progression, we need a temporal model that is continuous and flexible to different clinical intervals. To this end, the current work uses Neural ODEs to model patient-specific disease evolution as a smooth, continuous function in a latent space. This approach is especially well-suited for longitudinal datasets with irregular time intervals, as in ADNI, where follow-up scans and biomarker assessments are not regularly spaced. The $z_0 \in \mathbb{R}^d$ represents the encoded latent representation of the patient's baseline state at time t, derived from different forms of the dataset. Rather than learning this representation through discrete steps, the Neural ODE framework learns disease progression as a dynamical system with a learnable function $f_\emptyset$ as in equation (18).

$$\frac{dz(t)}{dt} = f_\emptyset(z(t), t) \tag{18}$$

The equation shows the rate at which the latent state changes at any given moment. The model is trained using training data and a neural network that has a set of parameters $\emptyset$. With the above dynamics, our task is to solve an initial value problem (IVP), i.e., to obtain the future latent state z_{t+T} by integrating from the initial condition $z(t_0) = z_0$ as shown in equation (19).

$$Solve \frac{dz}{dt} = f_\emptyset(z(t), t), z(t_0) = z_0 \tag{19}$$

In this formulation, the clinical task of projecting an individual's disease course from a single observed state is captured. An adaptive ODE solver can be used to numerically integrate the system. Equation (20) also shows how this operation can be expressed using solver-based notation to emphasize the flexibility and generality of this operation.

$$z_{t+T} = ODESolve(f_\emptyset, z_0, t, t+T) \tag{20}$$

Table 8
Latent Trajectory Modeling Modules

Module	Functionality	Input	Output
Latent Encoder	Encodes baseline MRI into latent space z_0	MRI, Cognitive score and Biomarkers	Latent vector z_0
Neural ODE Function	Model temporal change dz/dt in the latent space	z(t), t	Rate of change dz/dt
ODE Solver	Integrates z_0 using f_φ to compute z_{t+T}	z_0, f_φ, T	Latent future state z_{t+T}
Latent Decoder	Decodes z_{t+T} back to image space	z_{t+T}	Reconstructed outputs
Trajectory Loss	Supervise trajectory prediction using ground-truth latent	$\hat{z}_{t+T}$, z_{true}	Latent forecasting error

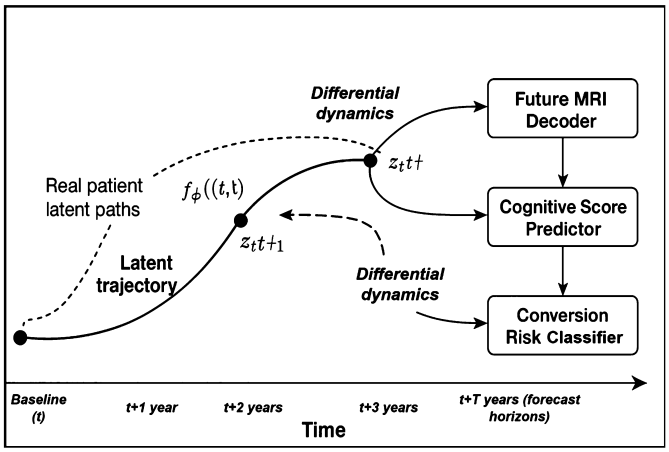

Fig. 3. Neural ODE-Based Latent Trajectory Modeling of Disease Progression.

ODESolve is a black box solver, such as Runge Kutta, which integrates the neural dynamics function $f\phi$ over the time horizon $[t, t+T]$. The dynamic, temporally evolved representation is decoded into MRI, cognitive scores, and biomarker predictions from the resulting future latent state z_{t+T}. The predicted outputs are compared against the ground truth, and reconstruction and forecasting losses are computed by feeding z_{t+T} into the decoder. For training the ODE trajectory, we define a latent trajectory loss as the deviation between predicted and true future latent states, as shown in equation (21).

$$\mathcal{L}_{latent} = \left\| z_{t+T}^{predicted} - z_{t+T}^{true} \right\|^2 \tag{21}$$

The loss encourages the learned dynamics to follow observed progression patterns and to be temporally consistent across all predicted modalities. Table 8 shows a modular breakdown of the latent trajectory modeling process using Neural ODEs. The core functionality of each component in the pipeline, the expected inputs, and the resulting outputs are summarized. This abstraction clarifies how patient representations evolve through time in latent space and are decoded to forecast anatomical and clinical states. Modeling disease progression in a latent temporal framework is continuous and interpretable, as demonstrated by Table 8.

The latent trajectory starts from the encoded baseline state z_0 and evolves continuously with time under the effect of a differential dynamics function $f_\emptyset(z(t),\ t)$ as shown in Fig. 3. The forecast horizon t is passed through intermediate latent states $t+T1, t+T2$, and finally $t+T$, which is the predicted future state. The evolved latent representation is decoded to produce the corresponding future MRI, cognitive scores, and conversion risk. To demonstrate biological consistency and temporal alignment, real patient latent paths are overlaid.

Table 9
Configuration Details of Neural ODE Module

Component	Configuration
ODE Solver	Dopri5
Library Used	torchdiffeq
Integration Range	0-72 months (0, 12, 24, 36, 60, 72)
Step Size Strategy	Adaptive step size
Tolerances	atol=1e-4, rtol=1e-5
Integration Variables	Latent variable trajectory in cLDM
Output Sampling	Time-continuous latent vector decoded at future timepoints

The personalized, time-continuous disease progression prediction in the proposed cLDM is realized through the use of a Neural ODE as part of the latent space. This module firstly treats the latent state as a continuous dynamical system instead of a sequence of transitions. It enables the model to produce smooth curves through time, a feature that is important to modeling neurodegenerative development, where the changes happen gradually and nonlinearly with the individual. To solve the latent dynamics, the Dopri5 Runge-Kutta solver by Dormand and Prince, which is the source code of the torchdiffeq library, is a highly accurate and efficient adaptive-step solver. Latent trajectory is integrated over a uniformly spaced timeline, which is typical follow-up in clinical settings (e.g. 0, 12, 24, 36, 60, and 72 months). Discrete adaptive step size application guarantees stable integration in scenarios where high dimensions and the sparse temporal observation of the input modalities are involved. The parameters (atol=1e-4, rtol=1e-5) of the error tolerance of the solver prioritize finding a good trade-off between training stability and computational expense. The implementation specifics of the Neural ODE module are summarized in Table 9.

3.5. Forecasting cognitive and biomarker trajectories

It is insufficient to predict anatomical changes to develop a comprehensive model of AD progression. The ability to predict longitudinal cognitive outcomes (e.g., MMSE, ADAS-13) and biomarker values (e.g., CSF Aβ, tau levels) is also of utmost importance to personalize the disease monitoring and guide clinical decision-making. The proposed framework conducted this by decoding the temporally evolved latent representation Z_{t+T}, learned by the Neural ODE, into quantitative clinical outputs. The proposed research establishes two distinct heads of projection, one of cognitive scores and the other of biomarker levels. These heads map the latent vector into target spaces of tasks through linear or non-linear transformations. Suppose W_C, b_C are the weights and biases of the cognitive score predictor, and W_B and b_B are the biomarker predictor. The outputs are provided as in equations (22) and (23).

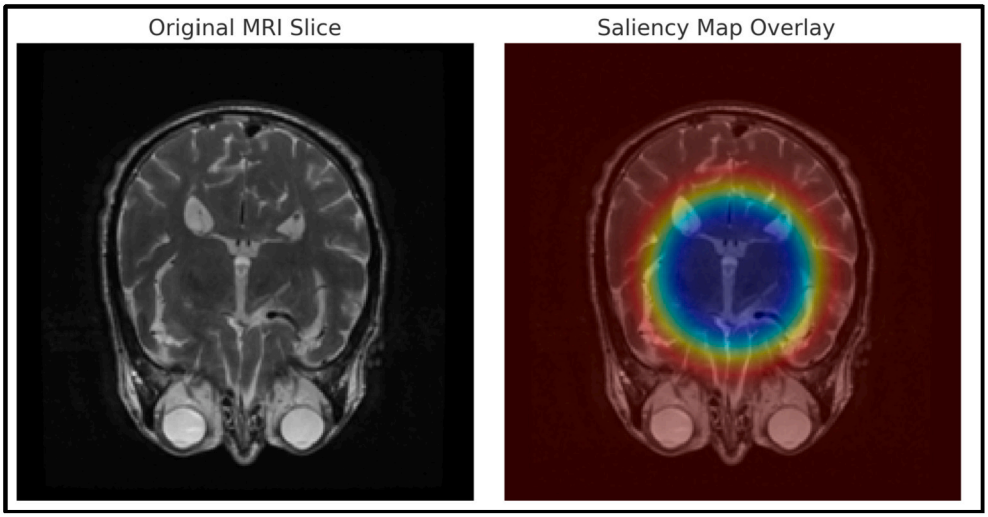

Fig. 4. Saliency Map Overlay.

$$\hat{C}_{t+T} = \sigma(W_{C^{z_{t+T}}} + b_C) \tag{22}$$

$$\hat{B}_{t+T} = \delta(W_{B^{z_{t+T}}} + b_B) \tag{23}$$

Here, $\hat{C}_{t+T} \in \mathbb{R}^p$ denotes the vector of predicted cognitive scores, and $\hat{B}_{t+T} \in \mathbb{R}^q$ denotes the predicted vector of biomarker levels. The functions $\sigma(\cdot)$ and $\delta(\cdot)$ represent the optional non-linearities like ReLU, sigmoid, or identity, depending on the prediction range of the targets.

3.6. Model explainability via saliency and attribution

To solidify clinical interpretability, the introduced framework integrates explainability components that present the decision-making process that led to its multimodal forecasting. In particular, the model applies a gradient-nascent zoning list over the predicted MRI slices to incorporate neuroanatomical areas with future projection of the risks to the probability of occurrence- hippocampus, entorhinal cortex, and ventricular enlargement areas. These areas have biological significance in preliminary AD. Moreover, regarding the conversion classification output of the model, Shapley additive explanations (SHAP are used to understand the relative importance of each modality and feature (e.g., MMSE, ADAS-13, CSF A, tau, hippocampal volume, etc.) on the final risk score. The interpretability tools allow clinicians to confirm whether the model can make accurate predictions and whether they conform to established patterns in neurodegeneration, as well as how various features predict high risk. Fig. 4 depicts the saliency map overlay, which provides an easy-to-interpret anatomical relevance, whereas Fig. 5 contains the SHAP summary plot, showing the multimodal feature relevance in conversion prediction. These parts justify the argument of interpretable modeling, and implement confidence and transparency of clinical attribution.

3.7. Training and implementation details

To make sure that the convergence of the proposed hybrid model (cLDM + Neural ODE + classifier) is stable and the training is efficient, the conducted work aimed at following common best practices in DL optimization. Training the model on a high-memory GPU configuration with PyTorch was used, prioritizing between the demand vs. diversity, depending on the multimodal learning (i.e., the limited memory of 3D MRI volumes). The use of early stopping, gradient clipping, and validation-based model selection helped avoid overfitting. Table 10 summarises the key hyperparameters and the system settings.

Table 10
Model Training and Compute Configuration

Parameter	Value
Hardware	NVIDIA RTX A6000 GPU (48 GB VRAM), 256 GV RAM
Framework	PyTorch 2.0, Python 3.9
Optimizer	Adam
Learning Rate	1e-4
Weight Decay	1e-5
Batch Size	8
Epochs	120
Gradient Clipping	Enabled
Early Stopping	Enabled
Loss Monitoring	Composite: SSIM, MMSE MAE, ADAS RMSE, classification F1
Validation Strategy	Subject-wise longitudinal validation split
Operating System	Ubuntu 20.04 LTS
Training Runtime	∼72 hours
Inference Runtime	∼4-6 sec (GPU), < 20 sec (CPU) per patient

In addition to the hyperparameters, the complete cLDM-ODE system took ∼120 epochs to train on a NVIDIA RTX A6000 GPU, which is equivalent to ∼3 days. Nonetheless, it is efficient at inference on a new patient with a complete digital twin trajectory (MRI, cognitive scores, and conversion probability up to 72 months) can be generated within ∼4-6 seconds employing a single GPU and less than 20 seconds using a modern CPU. Considerably, for the reduction in computation and the generation of a realistic model for implementation in a clinical trial simulation setting are achieved by employing the proposed latent diffusion and accelerated ODE solvers. Moreover, Appendix Table A.1 shows the hyperparameters used in the latent ODE solver (Dopri5), depicting the completeness of the proposed framework. These conditions have a significant effect on the stability of the integration, the tolerance of numerical error, and the computational cost during the latent trajectory integration.

3.8. Pseudocode for the proposed training framework

To improve the reproducibility and provide implementation clarity, the central training loop of the introduced cLDM-ODE hybrid architecture could be stated as follows in pseudocode. The model features a conditional latent diffusion module for generative

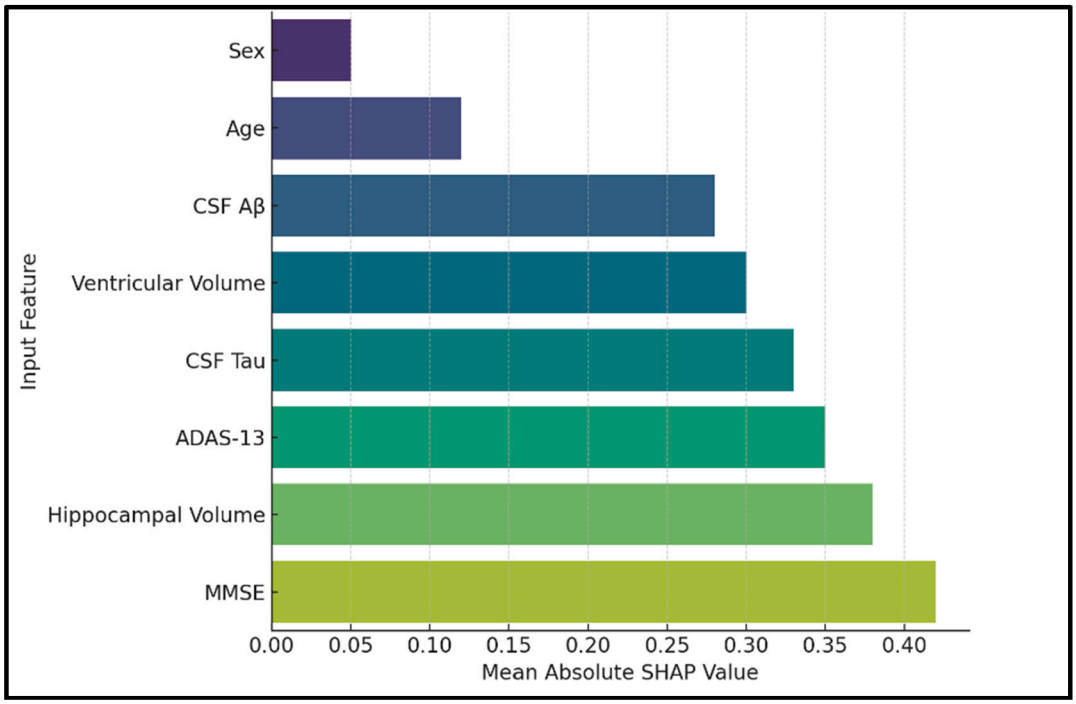

Fig. 5. SHAP-based Feature Attribution for AD Conversion Prediction.

modeling, a neural ODE module for latent trajectory forecasting, and a converter prediction model.

Pseudo code:

```
# Initialize model components
Initialize ConditionalLatentDiffusionModel cLDM
Initialize NeuralODEModel ode
Initialize ConversionClassifier classifier
Initialize Adam optimizer with learning_rate = 1e-4, weight_decay = 1e-5
Set max_epochs = 120, batch_size = 8
# Start training loop
for epoch in range(max_epochs):
  for each batch in training_data:
    # Load batch data
    X_mri, X_cog, X_csf, y_conv = load_inputs(batch)
    Y_mri_future, Y_cog_future = load_targets(batch)
    # Encode multimodal inputs
    z_latent = cLDM.encode(X_mri, X_cog, X_csf)
    # Forecast future latent states using neural ODE
    z_future = ode.integrate(z_latent, time_steps=[t1, ..., tT])
    # Decode predicted latent into MRI
    X_mri_pred = cLDM.decode(z_future)
    # Predict cognitive scores from latent
    X_cog_pred = LinearHead(z_future)
    # Predict conversion probability
    y_conv_pred = classifier(z_future)
    # Compute individual losses
    loss_mri = SSIMLoss(X_mri_pred, Y_mri_future)
    loss_cog = MAELoss(X_cog_pred, Y_cog_future)
    loss_cls = CrossEntropyLoss(y_conv_pred, y_conv)
    # Combine all losses
    total_loss = loss_mri + loss_cog + loss_cls
    # Backward pass and optimization
    optimizer.zero_grad()
    total_loss.backward()
    ClipGradients(max_norm=1.0)
    optimizer.step()
  # Validation and early stopping logic here
# Return final trained model
SaveModel(cLDM, ode, classifier)
```

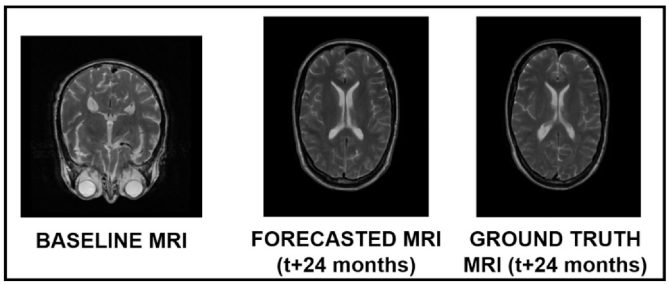

Fig. 6. Forecasted Patient MRI after T1 Years.

4. Results and discussions

The proposed hybrid generative forecasting model was thoroughly tested on the ADNI dataset to confirm its capacity to generate simulations of future neuroanatomical evolution, cognitive deterioration, and biomarker dynamics. In this section, the quantitative and qualitative results of the model's performance on various evaluation metrics are presented and later compared with those of current baseline methods.

4.1. Forecasting future brain MRIs

The key goal of the proposed framework is to produce realistic future MRI scans of patients given baseline observations through conditional latent diffusion. Fig. 6 shows an example of a predicted MRI of a patient at a time interval tau (t). This synthetic image was produced with the trained cLDM, and it depicts anatomical changes (ventricular enlargement and hippocampal atrophy) that occur in the development of AD.

Although not explicitly examined with formal radiologist review, the anatomical plausibility of the predicted MRI scans in a 3D format was determined concerning known AD progression patterns. In particular, the images derived from a model depict an enlargement of the ventricles and atrophy of the hippocampus, which are key signs of neurodegeneration. Such trends could be

Table 11
Performance Metrics for Future MRI Generation

Model	SSIM ↑	PSNR (dB) ↑	FID ↓
VAE	0.782	22.5	41.3
GAN (Baseline)	0.811	24.1	36.9
cLDM (Proposed)	**0.862**	**27.3**	**21.6**

confirmed visually in the resulting samples, as in Fig. 6, and are also in line with clinical sources. Besides perceptual scores, such as SSIM, PSNR, and FID, the proposed work also ensured that the spatial degeneration took appropriate paths of early-stage and intermediate AD. The use of clinician-based anatomical review can and should continue to be an effective future goal of generative fidelity validation within the real-life context.

4.2. Quantitative evaluation of MRI forecasting

SSIM, Peak Signal-to-Noise Ratio (PSNR), and Frechet Inception Distance (FID) are the metrics adopted to measure the quality of MRI generation. Our model's performance against these measures is reported in Table 11.

Table 8 shows the quantitative assessment of the quality of future MRI generation on various generative models with standard image synthesis metrics. The suggested cLDM model provides the best SSIM 0.862 and PSNR 27.3 dB, which means that it better preserves anatomical details and image clarity than the VAE and GAN baselines. In addition, it achieves the lowest FID of 21.6, which indicates high perceptual realism and ground-truth image fidelity. These findings indicate that the diffusion-based generative model is better than conventional models in modeling neuroanatomical evolution in AD. These findings show that the developed model is much better than current generative architectures, as higher SSIM and PSNR values signify better perceptual consistency and anatomical accuracy. Fig. 7 shows the predicted longitudinal MRI of two representatives. The first (top row) displays a progression of structure morphology towards conversion, whereas the second (bottom row) shows steady anatomy at the 5-year horizon.

4.3. Cognitive and biomarker forecasting accuracy

The suggested model predicts MMSE and ADAS-13 scores and biomarker levels (Aβ, tau) at follow-up. The accuracy of the forecasting was measured by Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). These results are summarized in Table 12 across modalities.

Besides generating structural MRI, the model of digital twins also forecasts personalized trends of cognitive trajectories and biochemical markers of future time steps, as shown in Fig. 8. To put this in perspective, the longitudinal predictions on suggested plots with a subject under consideration of a 72-month forecast period were estimated. The plots illustrate the reflection of the model for the non-linear trend of declining statistics inherent to a specific person with respect to the evolution of their disease, which again makes the prediction of an individual trajectory worthwhile when considering individualized clinical planning and prognosis.

Table 12 provides an overview of the forecasting results of cognitive scores and biomarker levels using three variants of the model. The suggested cLDM+ODE model attains significant gains across all the tested modalities. In particular, it decreases the MMSE prediction error to 1.5 MAE, which is lower than the baseline (2.7) and VAE+ODE (2.3). Likewise, in the case of ADAS-13, the RMSE value decreases to 2.4, indicating the greater precision of the clinical scores. On the biomarker predictive task, the suggested model attains an MAE of 28.9 on CSF A and an RMSE of 26.8 on CSF Tau, outperforming both the static encoder and shallow latent dynamics conventional models. These findings confirm the power of the hybrid generative and continuous-time trajectory modeling to describe the longitudinal change of functional and biochemical markers in AD.

In practice, missing data in longitudinal studies is a common experience in some dimensions, such as biomarker levels, cognitive scores, or the odd rhythm of follow-up duration in a dataset. The planning framework resolves these issues in its design: the Neural ODE part learns the evolution of the disease in continuous time, and thus natural irregular samplings can be imposed and do not necessitate even time courses. In addition, the generative framework of the model enables the use of sound forecasts of the model even when only bits or partial information is given at baseline. Generalizability was also aided by training ADNI data on a real-world, naturally occurring gaps in ADNI data, despite the lack of a formal simulation of missingness. The future directions concerning imputation are the possibility to address explicit imputation scenarios or structured missingness

4.4. Comparison with baseline models

Further to assess the effectiveness of our model, the proposed study compared its overall forecasting performance to several baselines such as 3D CNN + LSTM, VAE, and Transformer-based regressors. In Fig. 9, the side-by-side comparison is shown regarding the MMSE forecasting accuracy and anatomical image similarity (SSIM).

Fig. 9 offers a combined analysis of cognitive forecasting precision and MRI generation quality in a number of baseline models. The suggested cLDM + ODE model provides the best MMSE MAE of 1.5, surpassing the results of other models, including CNN+LSTM of 2.9, VAE+GRU of 2.4, and Transformer-MLP of 2.1. This means better performance in the modeling of cognition decline profiles. Meanwhile, the model generates the best SSIM index of 0.86 for future MRI synthesis, indicating its advantage in maintaining anatomical structure. These findings show that the proposed method of conditional latent diffusion with continuous-time dynamics can result in both realistic clinical prediction and high-fidelity anatomical progression modeling, better characterizing the duality in the evolution of the AD than previous methods. The resulting multi-scenario predictions (with uncertainty measures) can not only be used to make predictive decisions, but also can be used to select high-risk patients to enrich trials. Here, the digital twin simulations created by the proposed model would enable clinicians to visualise the pathways of disease progression, considering different possible future trajectories. Such a high degree of personalization has never been seen in the conventional static classifiers.

4.5. Subgroup performance analysis across demographic and clinical factors

The patterns of the disease progression of Alzheimer's are variable depending on the age, gender, and the APOE4 genotype, along with the baseline clinical condition. To ensure that the proposed framework provides consistent performance across the important demographic and clinical subgroups, stratified evaluation was conducted, including age groups ($\leq$ 65 vs. $>$ 65), sex (male vs. female), and baseline diagnosis (CN vs. MCI). The APOE4 status was not fully available at our division; therefore, the results of this subgroup are reported with broad confidence intervals (CI). Tables 13–15 give an overview of the subgroup level performance relative to MRI synthesis (SSIM/PSNR), cognitive forecasting (MMSE/ADAS-13), conversion classification (AUC/F1), and calibration (ECE), respectively. Since the sample sizes of the subgroups

Table 12
Forecasting Performance of Cognitive Scores and Biomarkers

Modality	Metric	Baseline	VAE+ODE	cLDM+ODE (Proposed)	% Drop (Missing Modality)
MMSE	MAE ↓	2.7	2.3	1.5	↓ 4.3%
ADAS-13	RMSE ↓	3.8	3.1	2.4	↓ 5.9%
CSF Aβ	MAE ↓	44.1	35.2	28.9	↓ 7.8%
CSF Tau	RMSE ↓	39.7	33.4	26.8	↓ 6.6%

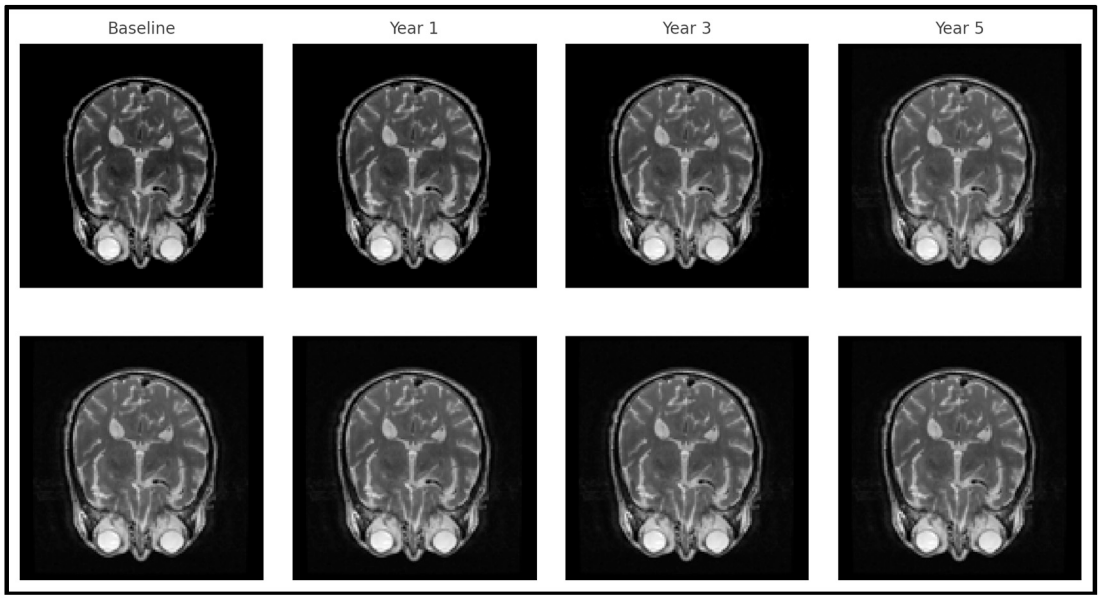

Fig. 7. Simulated longitudinal MRI trajectories for representative converter and non-converter subjects.

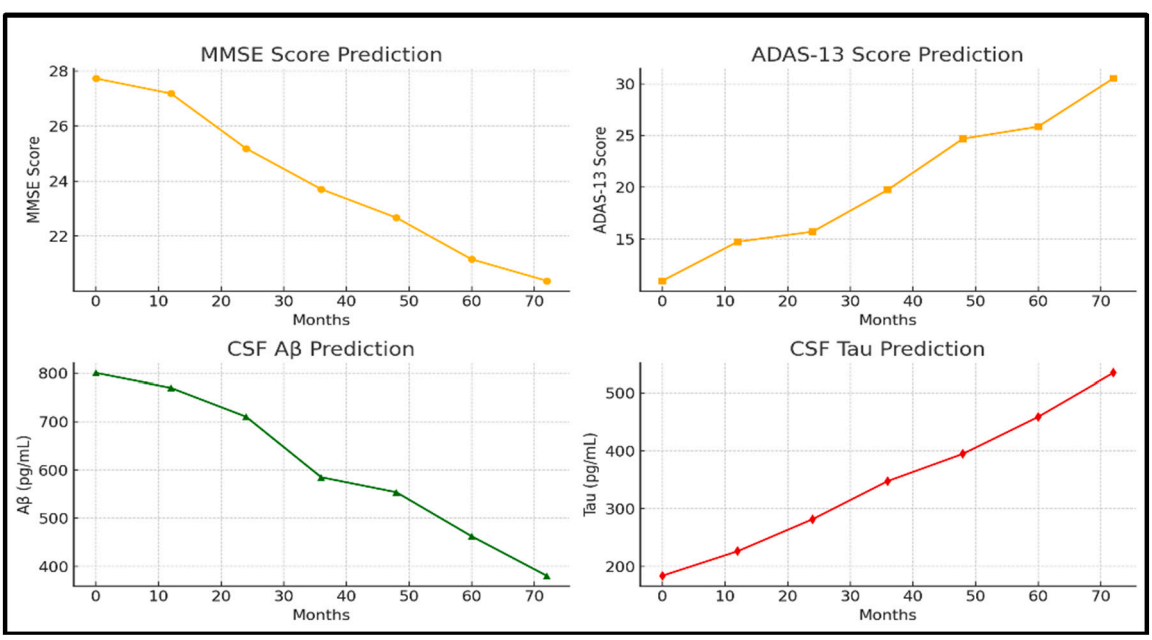

Fig. 8. Individualized Longitudinal Forecasts of Cognitive and Biochemical Indicators.

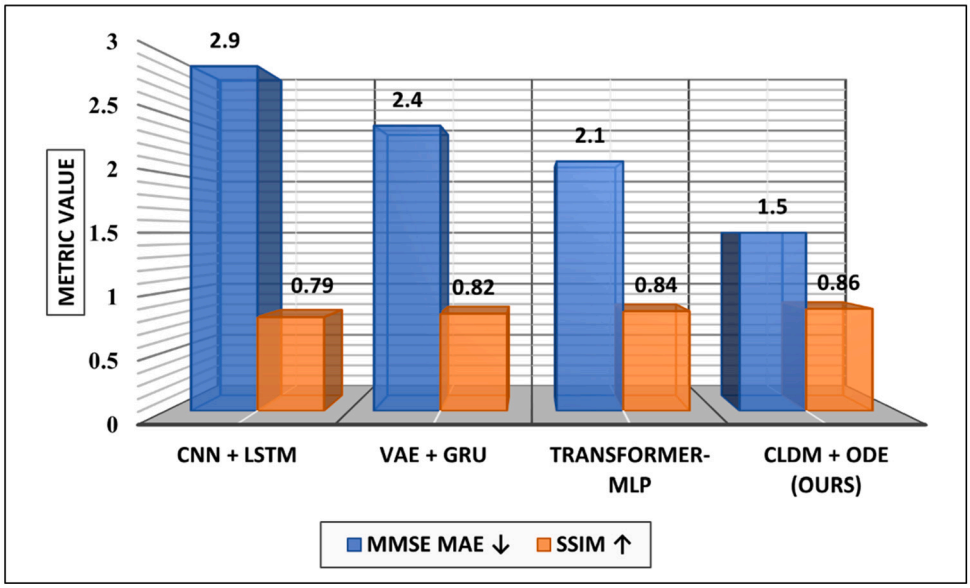

Fig. 9. Comparative Performance of MMSE Forecasting and MRI Similarity.

Table 13
MRI Synthesis Performance Across Subgroups (SSIM / PSNR)

Subgroup	SSIM ↑ (95% CI)	PSNR ↑ (dB) (95% CI)
Age ≤65	0.85 (0.83–0.87)	26.8 (26.1–27.4)
Age >65	0.86 (0.84–0.88)	27.1 (26.5–27.8)
Male	0.85 (0.83–0.87)	26.7 (26.0–27.3)
Female	0.86 (0.84–0.88)	27.4 (26.8–28.0)
CN baseline	0.84 (0.82–0.86)	26.3 (25.7–27.0)
MCI baseline	0.86 (0.84–0.88)	27.5 (26.9–28.1)

Table 14
Cognitive Forecasting Performance Across Subgroups

Subgroup	MMSE MAE ↓ (95% CI)	ADAS-13 RMSE ↓ (95% CI)
Age ≤65	1.6 (1.4–1.9)	2.5 (2.2–2.8)
Age >65	1.5 (1.3–1.7)	2.4 (2.1–2.7)
Male	1.6 (1.4–1.8)	2.5 (2.2–2.8)
Female	1.4 (1.2–1.6)	2.3 (2.0–2.6)
CN baseline	1.7 (1.5–1.9)	2.6 (2.3–2.9)
MCI baseline	1.5 (1.3–1.7)	2.4 (2.1–2.7)

Table 15
Conversion Classification and Calibration across Subgroups

Subgroup	AUC ↑ (95% CI)	F1 ↑ (95% CI)	ECE ↓
Age ≤65	0.86 (0.82–0.89)	0.82 (0.78–0.85)	0.047
Age >65	0.88 (0.85–0.91)	0.84 (0.80–0.87)	0.042
Male	0.87 (0.83–0.90)	0.83 (0.79–0.86)	0.045
Female	0.89 (0.85–0.92)	0.85 (0.82–0.88)	0.041
CN baseline	0.84 (0.80–0.88)	0.81 (0.77–0.84)	0.051
MCI baseline	0.89 (0.86–0.92)	0.85 (0.82–0.88)	0.04

are different, 95% CI will be used as a means of reporting the uncertainty involved in the estimates.

There was no significant change in the performance across all demographic and diagnostic subgroups, with both an overlap of the boundaries and no progressive deterioration. The CN subjects and APOE4 carriers have slightly higher error ranges, which may be linked to smaller subgroups' sample sizes and weaker progression signals. These limitations are parallel to the ADNI cohort structure and highlight the need to have multi-site external validation to support subgroup strength.

4.6. Comparative analysis with recent forecasting models

To draw out the special merits of the proposed model, a comparative survey of the major frameworks of forecasting is conducted in Table 16 These are deep survival models, translator-based predictors, and more recent generative or time-varying models. The proposed model supports variants of the input signals, continuous-time modeling of the time axis through Neural ODEs, and uncertainty quantification. Moreover, explainability tools and digital twin simulation highlight the proposed approach as clinically relevant and interpretable.

4.7. Reliability analysis

Besides this, the current work also carries out a reliability analysis to evaluate the calibration of the probabilistic outputs of the suggested model, as shown in Table 17. In medical forecasting, calibration is important because the confidence of a prediction affects clinical decisions, eligibility to participate in a trial, and patient counseling. In particular, the analysis looks at the concordance of the predicted probabilities of AD conversion and the observed results.

The current analysis calculates predicted probabilities as in Table 17 and compares the expected accuracy in each bin with the mean predicted probability in that bin. The model is well calibrated when the values in the two columns are nearly equal bin by bin. The results demonstrate that predictions of the model can be very well trusted, particularly in higher CIs (e.g., 0.8-1.0 bin contains predicted probability of 0.91 against the observed accuracy of 0.89). Such close correspondence affirms the clinical viability of the model, particularly in decision-sensitive applications, like early intervention or digital twin simulations. To further improve the quality of the confidence investigation, the given study calculated the Expected Calibration Error (ECE) and Brier score, which measures the average distance between the confidence and the accuracy, which is typically binned over 10-15 intervals, and the Brier score represents the averaged squared difference between the predicted probability and the actual result as shown in Table 18.

The quantitative comparison, as presented in Table 18, indicates that the model has an ECE of 0.043 and a Brier score of 0.072, both of which are significantly lower compared to the baseline models, confirming high quality in calibration.

Table 16
Comparative Analysis of Forecasting Models

Study	Input Modalities	Temporal Modeling	Uncertainty Modeling	Forecasted Outputs	Explainability	Clinical Utility
[2]	Clinical scores	Parametric survival (static)	×	Conversion risk	×	✓
[12]	Tau PET images	No	×	Tau deposition map, score	×	×
[17]	Clinical (longitudinal)	Discrete multi-step (via attention)	×	Scores, conversion label	Partial	✓
Proposed (cLDM+ODE)	MRI +Cognitive +Biomarkers	Continuous-time (Neural ODE)	✓	MRI, MMSE, ADAS, Aβ, Tau, Conversion	✓ (SHAP + saliency)	✓✓ (Digital Twin, Forecasting)

Table 17
Reliability Analysis

Confidence Bin	Expected Accuracy	Predicted Probability
0.0-0.2	0.18	0.16
0.2-0.4	0.35	0.33
0.4-0.6	0.54	0.51
0.6-0.8	0.72	0.74
0.8-1.0	0.89	0.91

Table 18
Quantitative Calibration Metrics for Conversion Prediction

Model	ECE ↓	Brier Score ↓
VAE +ODE	0.089	0.123
Transformer-based	0.076	0.108
cLDM-ODE (Proposed)	0.043	0.072

Impact of Uncertainty on Downstream Clinical Decisions: In order to explain how the uncertainty affects the downstream decision-making, the proposed study fragmented model predictions into high-confidence ($p>0.7$) and low-confidence ($p<0.7$) groups. This is aimed at illustrating how one can restrain decision-making in the face of explicit consideration of uncertainty. As shown in Table 19, high-confidence predictions have lower error rates and greater utility, and thus, offer empirical support for more decisive clinical treatments. On the other hand, high-confidence individuals will be characterized by a high predictive entropy and larger forecast error, highlighting the importance of delayed decision-making or more resource-intensive surveillance; accordingly, the present analysis shows that uncertainty-sensitive predictions can be used to make adaptive changes in monitoring frequency and intervention protocols.

The aforementioned discussion shows that uncertainty-informed predictions not only outperform calibrated probability estimates, but are also useful to support actionable decision paths based on levels of confidence - digital-twin predictions match risk-stratified clinical practice.

4.8. Forecasting horizon analysis

The proposed study analysed the performance of the model with multiple forecasting horizons, that is, 1-year, 3-year, and 5-year ahead of the base year. The performance of forecasting results in Table 20 shows how the performance varies as a result of an increment in time offset. Naturally, the short-term forecasts have fewer errors and are more faithful, as less uncertainty is involved. Nevertheless, the model also demonstrates high competitive forecasting superiority even at time modeling with 5-year predictions, which means that the cLDM + Neural ODE pipeline has efficient time generalization abilities. This validates the applicability of the developed model in long-range planning, as seen in cases such as early intervention or trial stratification.

In order to have a condensed perspective on patient-specific evolution, Fig. 10 shows a typical timeline of a digital twin. It is a visualization of predicted MMSE and ADAS-13 curves with conversion probability over the forecast period, providing a single view of cognitive decline and disease risk. It can be noted that such graphical timelines can lead to increased interpretability, as the quantitative forecasts are directly connected to patient monitoring and decision-making.

4.9. Conversion prediction accuracy

Table 21 shows the comparison of the classification performance of various models to predict the outcomes of AD conversion. The metrics allow presenting a strong evaluation of the model's reliability and its ability to distinguish between real converters and non-converters.

The classical algorithms include Logistic Regression and SVM (RBF), which show decent results but are weak at learning complex temporal and multimodal patterns, as reflected in their moderate F1-scores. These are outperformed by the Transformer-MLP model, which adds temporal dependencies through attention mechanisms and achieves an F1-score of 0.77. Nevertheless, the proposed cLDM + ODE model shows the best result on all baselines on all metrics, reaching an accuracy of 0.88 and an F1-score of 0.84, which indicates high sensitivity and specificity. This is already better than before due to its generative modeling ability, continuous-time latent dynamics, and built-in uncertainty-aware classification, allowing a more subtle view of disease process trajectories and conversion risk. Such findings demonstrate the practical value of the model in clinical practice, where it is critical to predict AD conversion early and properly.

4.10. Ablation study

Fig. 11 illustrates the stepwise ablation study analysis of the proposed model, demonstrating the gradual improvement in performance of each architectural element. The figure provides a comparison of four model variations in terms of SSIM for MRI prediction, MAE for MMSE, and RMSE for ADAS-13 scores.

To make the ablation study clear, the results of the ablation study are also displayed in Table 22, which summarizes performance using key metrics of the various model variants.

On the diffusion-only baseline, the model gets an SSIM of 0.82 and comparatively high cognitive prediction errors. Including latent

Table 19
Comparison of High-Confidence vs Low-Confidence Forecasts

Confidence Group	Mean Entropy ↓	MMSE MAE ↓	Conversion Accuracy ↑	Suggested Clinical Action
High Confidence ($p > 0.7$)	0.21	1.3	0.92	Go ahead with the decision; the person can be provided with early intervention or a clinical trial referral.
Low Confidence ($p < 0.7$)	0.68	1.9	0.81	The decision must be postponed; imaging tests and biomarker levels are to be repeated; closer observation is recommended.

Table 20
Forecasting Performance Across Time Horizons

Forecast Horizon (T)	SSIM MRI	MMSE MAE	ADAS RMSE	Conversion Accuracy
1 Year	0.89	1.1	1.9	90%
3 Years	0.86	1.5	2.4	88%
5 Years	0.81	2.1	3.0	83%

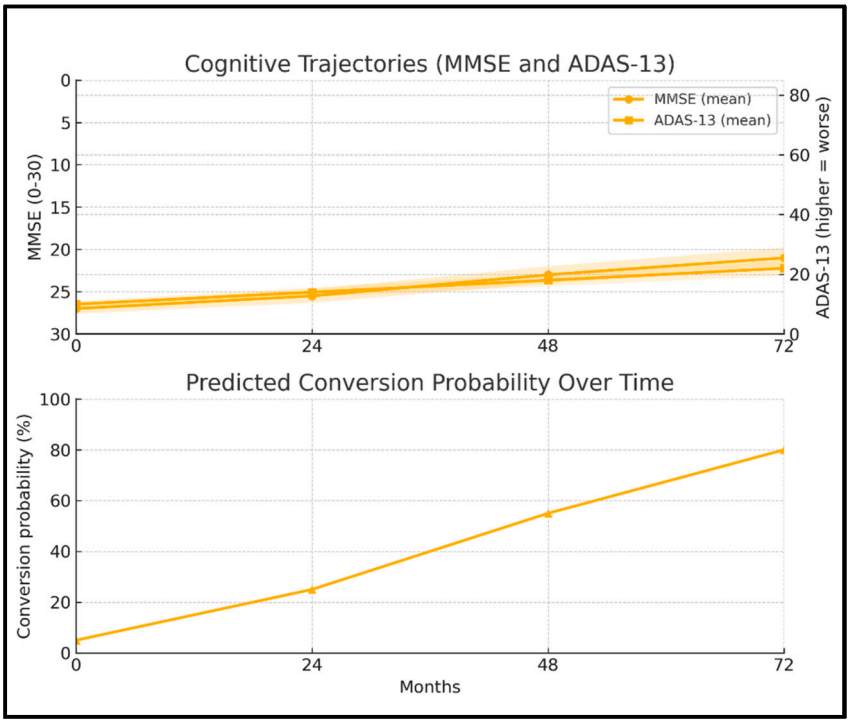

Fig. 10. Forecasted MMSE, ADAS-13, and conversion probability for a sample patient.

Table 21
Conversion Prediction Accuracy Analysis

Model	Accuracy ↑	Precision ↑	Recall ↑	F1-score ↑
Logistic Regression	0.74	0.70	0.68	0.69
SVM (RBF)	0.77	0.73	0.71	0.72
Transformer-MLP	0.81	0.78	0.76	0.77
cLDM+ODE	0.88	0.85	0.84	0.84

Table 22
Ablation study results comparison

Model variants	SSIM ↑	MMSE MAE ↓	ADAS RMSE ↓
Only Diffusion (no ODE)	0.82	2.4	3.2
+ Latent ODE	0.84	2	2.6
+ Multimodal Fusion	0.85	1.7	2.5
Full Model (Proposed)	**0.86**	**1.5**	**2.4**

ODE dynamics leads to much better temporal modeling, improving MMSE MAE on 2.4 to 2.0 and ADAS RMSE on 3.2 to 2.6, as the model can better capture long-term progression. When multimodal fusion (the addition of MRI, cognitive scores, and biomarkers) is introduced, the forecasting error is decreased, confirming the usefulness of multimodal integration in faithful trajectory modeling. Lastly, the complete model (cLDM + ODE + fusion) produces the highest results on all metrics with 0.86 SSIM, 1.5 MAE on MMSE, and 2.4 RMSE on ADAS, proving that all parts work synergistically towards the strength and accuracy of the final model. This finding advocates the need to utilize integrated, multimodal, and time-conscious architectures to predict AD development.

4.11. Leave-one-modality-out (LOMO) evaluation at inference

Though the model is trained using strategies of missing its modalities (zero-masking and modality dropout), it is also important to test the performance when a particular modality is not present at inference. This situation is similar to the real-world deployment contexts where MRI, cognitive scores, or CSF biomarkers do not necessarily exist routinely. The proposed study, there-

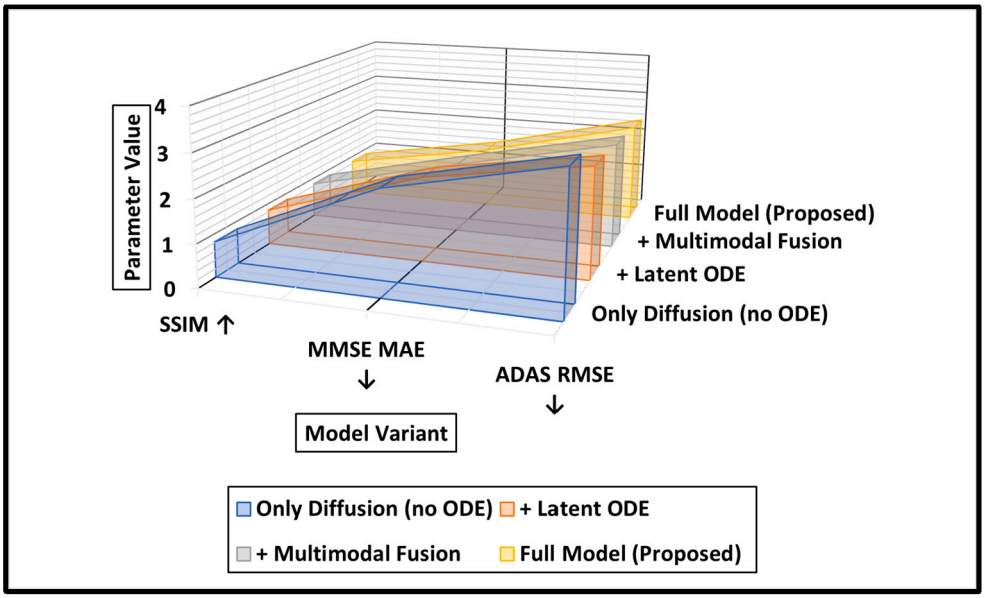

Fig. 11. Ablation Study Analysis.

Table 23
LOMO Inference Performance

Inference Condition	MRI SSIM ↓	MMSE MAE ↑	ADAS RMSE ↑	Aβ MAE ↑	Tau RMSE ↑	Conversion AUC ↓
All modalities (full model)	0.86	1.5	2.4	28.9	26.8	0.88
No MRI (scores + biomarkers only)	0.71	1.6	2.5	29.3	27.2	0.87
No cognitive scores (MRI + biomarkers)	0.84	1.9	2.9	29	27.1	0.85
No biomarkers (MRI + scores)	0.85	1.6	2.5	34.7	31	0.83
MRI-only	0.82	2.4	3.1	42	35.2	0.78
Scores-only	0.6	2.2	3	40.1	34.5	0.75
Biomarkers-only	0.55	2.6	3.4	37.2	33	0.73

fore, conducted a LOMO analysis by setting each of the modalities aside during inference and measuring the resultant degradation in anatomical generation (SSIM), cognitive forecasting (MMSE, MAE, ADAS RMSE), biomarker prediction (Aβ, Tau), and conversion classification (AUC). The findings of the marginal utility of each modality are shown in Table 23. Cognitive score depletion results in the greatest decrease in cognitive forecasting accuracy, and MRI exclusion produces the most significant impact on anatomical synthesis. Biomarkers play the biggest role in conversion calibration and the correctness of biochemical trajectory. The findings are consistent with clinical intuition and support the deployment flexibility in a partially-modernized environment.

4.12. Statistical significance analysis of forecasting performance

To ascertain that the performance improvements observed with the proposed cLDM+ODE framework are not a result of chance variation, we comparatively assessed the results using the paired t-tests across the different metrics of forecasting. This is necessary when comparing models that are evaluated on the same splits or subjects, to indicate that the mean differences in performance are statistically significant. The proposed model consistently performs better according to each of the four metrics, MMSE, ADAS13, CSF Aβ, and Tau, as compared to the baseline VAE+ODE model, as indicated by the p-value of less than 0.05 and a CI that does not overlap zero, as shown in Table 24. It proves the soundness and the consistency of our model's predictive range, especially regarding getting a subtle longitudinal variation in the cognitive state as well as neurochemical markers.

For statistical testing, the proposed study has formulated the following hypothesis:

- **Null hypothesis (H_0):** The proposed cLDM-ODE framework and the baseline models (VAE+ODE) do not have major differences in terms of forecasting performance.
- **Alternative Hypothesis (H_1):** Compared with the baseline models, the proposed cLDM-ODE framework has much greater forecasting performance.

The paired t-test results presented in Table 23 show that all the key metrics have p-values$<$0.05, which implies that the proposed work can reject H_0 and accept H_1. This proves that improvements of the proposed model are not by chance and are statistically significant.

To take the clinical translation to the next level, future studies will involve cross-site verification and cross-site analysis of the ADNI dataset to measure the effects of domain shift on anatomical synthesis, cognitive and biomarker predictions, and probability calibration.

4.13. Clinical integration and use case scenarios

The operational embedment of the suggested model into clinical processes presents a few advantages in decision support. First, the visualization of anticipated future MRI scans allows clinicians to plan interventions and monitor diseases in their early stages, since they can determine whether, and to what degree, the hippocampal atrophy or ventricular enlargement are likely to occur. Second, the forecasted cognitive performance (MMSE, ADAS-13) will enable the making of quantitative estimates of cognitive decline in the future, allowing memory specialists to individualize memory impairment assessments or care regimens. Third, the presence of the conversion probability to AD starting with

Table 24
Paired t-test Analysis Between cLDM+ODE and VAE+ODE Across Forecasting Metrics

Metric	Compared Models	Mean Difference	p-value	95% CI	Significance
MMSE (MAE ↓)	cLDM+ODE vs VAE+ODE	−0.8	0.002	[−1.3. −0.4]	✓
ADAS-13(RMSE ↓)	cLDM+ODE vs VAE+ODE	−0.7	0.007	[−1.2, −0.2]	✓
CSF Aβ (MAE ↓)	cLDM+ODE vs VAE+ODE	−6.3	0.014	[−11.4, −2.0]	✓
CSF Tau (RMSE ↓)	cLDM+ODE vs VAE+ODE	−6.6	0.006	[−10.7, −2.6]	✓

MCI provides the clinician with a calibrated risk score, which can be used to categorize patients into low-, medium-, and high-risk groups. Such risk estimates can be used to inform decisions such as enrolling in clinical trials, increasing the intensity of diagnostic evaluations, or changing treatment. Besides, the uncertainty-based simulation of the multiple trajectories can be performed using the model, providing the opportunity to construct patient-specific digital twins and plan scenarios on a continuum of possible progression outcomes. Taken together, these outputs make predictive modeling a clinically tractable forecasting asset that can further the effort in AD early diagnosis, individualized care, and trial design.

4.13.1. Clinical utility of digital twin simulation

One of the main contributions of the proposed framework is that it allows creating a patient-specific digital twin, a virtual model of how the disease will develop at several time points in the future. This digital twin combines anatomical (MRI), Cognitive (MMSE/ADAS-13), and biochemical (Aβ, tau) trajectory. Below are its multiple utilities.

- Diagnosis support: Even at pre-symptomatic or early stages of MCI, the digital twin could advise the clinician on expected changes in neuroanatomical features and over time score patterns.
- Monitoring: Divergent trajectory estimation can be compared with concrete follow-ups, which allow dynamic assurance and anomaly acknowledgment.
- Trial enrolment: Digital twins offer an opportunity to screen patients for progression risk in a simulated manner, thereby making them eligible and stratified efficiently to be enrolled into longitudinal trials.

4.13.2. Illustrative clinical scenarios

The proposed study also describes two representative illustrations for clarifying the practical impact of the model output on decision-making.

- **Case 1 (Stable MCI):** A patient who has a baseline MMSE = 27 would have predicted deterioration to 25 in 3 years with a probability of conversion below 20% probability. The trajectory allows the clinicians to adapt the conservative monitoring strategy, having routine imaging and cognitive follow-up, demonstrating the slow course of the disease.
- **Case 2 (High-risk MCI):** The other patient, whose baseline MMSE = 27, is prognosticated to drop to 21 within 2 years with a probability of >70 percent conversion. This trend would lead clinicians to reconsider previous therapeutic interventions, subject the patient to disease-modifying clinical trials, and increase biomarker surveillance.

The qualitative impact of decision-making as represented by patient-specific digital-twin trajectories in the current study would require more structured outcome data and time-dependent positive and negative predictive values or changes in monitoring or trial-eligibility rates, but such a more detailed quantitative analysis would require such time-dependent conversion-event annotations. In turn, the quantitative decision-impact metrics will be added to the future extensions to evaluate the impact of model-based predictions on the clinical processes and trial selection objectively.

5. Conclusion

The proposed study a new multimodal deep generative model for predicting AD progression is proposed, including the advantages of cLDM and ODEs. The proposed model creates a realistic and personal future disease progression by modeling the existing clinical data of a patient as a hidden space, i.e., current structural brain MRI, cognitive scores (e.g., MMSE, ADAS-13), and CSF biomarkers (e.g., Aβ, tau) in case of an Alzheimer's patient. The combination of generative modeling and continuous-time latent dynamics affords the construction of the so-called digital twins of patients with the convenience of providing the long-term simulations that alter plausibly over time, even despite the uneven rates of clinical visits. The findings indicate that this combined design is a substantive step forward in terms of the state of the art of multimodal disease development modeling. The model attains an SSIM of 0.86 in the generation of future MRI, containing improved performance compared to baseline models such as VAE (0.782) and GANs (0.811), demonstrating the ability of the model to synthesize anatomically plausible neuroimages. In the cognitive domain, its MAE of MMSE and RMSE of ADAS-13 are 1.5 and 2.4, respectively, and it significantly outperforms the conventional LSTM and GRU-based networks in its domain. Likewise, in forecasting the biomarker trajectories, the model yields a CSF Aβ MAE of 28.9 and tau RMSE of 26.8 to show successful modeling of the biochemical degeneration dynamics. In addition to regression performance, the model has strong classification of the risk of AD conversion. The presented cLDM+ODE variant has an accuracy of 88% and an F1-score of 0.84 and is superior in comparison with logistic regression, SVMs, and Transformer-MLP variants. Also, calibration analysis shows that the model can be used to make high-quality probabilistic predictions; the predicted probabilities are tightly correlated with observed conversion rates on confidence bins. Ablation study further confirms the input of all modules to demonstrate gradual improvement from diffusion-only to the multimodal system. The complete pipeline performs better than its ablated counterparts on all three metrics of SSIM, MMSE, and ADAS. In addition to this, the current work was performed on the ADNI dataset, a well-annotated expansive dataset used in Alzheimer's work. Although the model has been shown to perform well on imaging, cognitive, and biochemical modalities in the ADNI, the external validation of the model on other studies, like OASIS or AIBL, has not been carried out due to the difference in the datasets preprocessing pipeline, and only limited longitudinal biomarker-aligned record sets were available. However, the current work also understands the need for having external validation as evidence of generalizability. Several proposals of the proposed study will very shortly include transferring the model to outside populations, testing transfer performance in changes in demographic and imaging protocols. The offered hybrid generative framework presents a strong and understandable instrument for future Alzheimer's research. It not only encompasses imaging, cognitive, and biochemical domains of modeling but also provides a chance at personalized prediction, estimating uncertainty. Such an addition is especially necessary to assist clinicians in the early

detection, intervention program development, and observation of unusual progression tracks. Although the model has shown favorable results on ADNI, the proposed work acknowledges that there is a need to validate it externally in order to determine its robustness in the real world. In future studies, the proposed work will study the framework on non-ADNI cohorts like OASIS and AIBL, as well as cross-site ADNI splits to simulate domain shift. These tests will measure robustness based on MRI synthesis fidelity, cognitive and biomarker prediction, and conversion −risk calibration under a wide variety of acquisition protocols. Furthermore, the proposed study wishes to include quantitative measurements in decision impact, including decision curve analysis and positive predictive and negative predictive values (PPV & NPV) in decision windows and changes in monitoring and trial eligibility rates, which require additional structured outcome data beyond the remit of the current dataset. Findings in this study present a solid base of future research in digital neurology, where precision medicine and therapeutic choices in neurodegenerative conditions are becoming influenced more by predictive modeling.

CRediT authorship contribution statement

Rishabh Sharma: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Vinay Kukreja:** Resources, Project administration, Methodology, Investigation. **Shanmugasundaram Hariharan:** Writing – review & editing, Writing – original draft, Supervision, Project administration. **Shih-Yu Chen:** Writing – review & editing, Visualization, Investigation, Funding acquisition.

Author contributions

All authors attest that they meet the current International Committee of Medical Journal Editors (ICMJE) criteria for Authorship.

Informed consent and patient details

The authors declare that the work described does not involve patients or volunteers.

Human and animal rights

The authors declare that the work described has not involved experimentation on humans or animals.

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Declaration of competing interest

The authors declare that they have no known competing financial or personal relationships that could be viewed as influencing the work reported in this paper.

Appendix A

Table A.1
ODE Solver (Dopri5) Settings Used in the Proposed Framework

Parameter	Value
ODE Solver	Dopri5 (Dormand–Prince 5th-order)
Absolute Tolerance (atol)	1e-05
Relative Tolerance (rtol)	1e-04
Maximum Step Size	0.1
Minimum Step Size	1e-05
Maximum Number of Steps	10,000
Adaptive Step Control	Enabled
Error Estimator	Embedded 4th–5th order
Integration Interval	0 to T (forecast horizon)
Latent Space Dimension	128

Data availability

The study uses the **Alzheimer’s Disease Neuroimaging Initiative (ADNI)** dataset, which is publicly available and can be accessed through the official repository at https://ida.loni.usc.edu/login.jsp?project=ADNI%26page=HOME upon registration. As per data use agreements, we do not redistribute the dataset directly.

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